Concavity, Curve Sketching, and Optimization

ANSWER KEY

Increasing/decreasing and the first derivative test

1 For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. $f' > 0$ on $(-\infty, 1)$, $f' < 0$ on $(1, 3)$, $f' > 0$ on $(3, \infty)$. So f is increasing on $(-\infty, 1) \cup (3, \infty)$, decreasing on $(1, 3)$. $x = 1$ is a local max (sign $+ \to -$); $x = 3$ is a local min (sign $- \to +$).

Concavity and the second derivative test

2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

$f'(x) = 4x^3 - 12x^2$; $f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ on $(-\infty, 0)$, $f'' < 0$ on $(0, 2)$, $f'' > 0$ on $(2, \infty)$. Concave up on $(-\infty, 0) \cup (2, \infty)$, concave down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$ (both sign changes).

3 Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 3x$.

$f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$. $f''(x) = 6x$. $f''(1) = 6 > 0$: local min at $x = 1$. $f''(-1) = -6 < 0$: local max at $x = -1$.

Curve sketching

4 For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.

$f'(x) = 3x^2 - 6x = 3x(x-2)$: critical points $x = 0, 2$. $f''(x) = 6x - 6$: zero at $x = 1$. Sign chart: $f' > 0$ on $(-\infty, 0)$, $f' < 0$ on $(0, 2)$, $f' > 0$ on $(2, \infty)$ — local max at $x = 0$ ($f(0) = 0$), local min at $x = 2$ ($f(2) = -4$). $f'' < 0$ on $(-\infty, 1)$ (concave down), $f'' > 0$ on $(1, \infty)$ (concave up) — inflection point at $x = 1$ ($f(1) = -2$).

Optimization problems

5 A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.

Let x = side parallel to river, y = each perpendicular side. Constraint: $x + 2y = 200$, so $x = 200 - 2y$. Area $A = xy = (200 - 2y)y = 200y - 2y^2$. $A'(y) = 200 - 4y = 0 \Rightarrow y = 50$. $x = 200 - 100 = 100$. Maximum area occurs at $x = 100$ m, $y = 50$ m (verified max since $A'' = -4 < 0$).

6 An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.

$V(x) = x(12-2x)^2$ for $0 < x < 6$.
$V'(x) = (12-2x)^2 + x \cdot 2(12-2x)(-2) = (12-2x)[(12-2x)-4x] = (12-2x)(12-6x)$. Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. $V(2) = 2(8)^2 = 128$ in^3, the maximum.

Optimization: minimizing cost or distance

7 Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

$D(x) = x^2 + (2x-3)^2 = x^2 + 4x^2 - 12x + 9 = 5x^2 - 12x + 9$. $D'(x) = 10x - 12 = 0 \Rightarrow x = \frac{6}{5}$.
$y = 2\left(\frac{6}{5}\right) - 3 = \frac{12}{5} - 3 = -\frac{3}{5}$. Closest point: $\left(\frac{6}{5}, -\frac{3}{5}\right)$ (confirmed min since $D'' = 10 > 0$).

8 A closed cylindrical can must have volume 16π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes surface area.

From V: $h = \frac{16}{r^2}$. $S(r) = 2\pi r^2 + 2\pi r \cdot \frac{16}{r^2} = 2\pi r^2 + \frac{32\pi}{r}$. $S'(r) = 4\pi r - \frac{32\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{32\pi}{r^2} \Rightarrow r^3 = 8 \Rightarrow r = 2$ cm.

Points of inflection and graph behavior

9 Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

$f'(x) = 3x^2 \geq 0$ everywhere, so f is never decreasing — $x = 0$ is not a local extremum. However $f''(x) = 6x$ changes sign at $x = 0$ (from concave down to concave up), so $x = 0$ IS an inflection point. This shows concavity change and extrema are independent features.

Visualizing extrema and concavity together

10 The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.

The graph rises to a peak at $x = 0$ ($f = 0$), then falls, changing its bend from concave-down to concave-up right at $x = 1$ ($f = -2$, the inflection point), before reaching a valley at $x = 2$ ($f = -4$) and rising again — visually confirming all three features found algebraically.

A capstone optimization problem

11 A rectangular box with a square base and no top must have volume 500 cm^3. Using $V = x^2 h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.
$S(x) = x^2 + 4x \cdot \frac{500}{x^2} = x^2 + \frac{2000}{x}$. $S'(x) = 2x - \frac{2000}{x^2} = 0 \Rightarrow 2x^3 = 2000 \Rightarrow x^3 = 1000 \Rightarrow x = 10$ cm.

Second derivative test failures

12 For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.

$f'(x) = 4x^3 = 0$ gives $x = 0$. $f''(x) = 12x^2$; $f''(0) = 0$, so the Second Derivative Test gives no information. Using the First Derivative Test: $f'(x) < 0$ for $x < 0$ and $f'(x) > 0$ for $x > 0$, so $x = 0$ is a local (and in fact absolute) minimum.

A capstone curve-sketching problem

13 For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2) = 12x(x - 2)(x + 1)$: critical points $x = -1, 0, 2$.
$f''(x) = 36x^2 - 24x - 24$. $f''(-1) = 36 + 24 - 24 = 36 > 0$: local min. $f''(0) = -24 < 0$: local max.
$f''(2) = 144 - 48 - 24 = 72 > 0$: local min.